



Pittala Sree Sai Pavan

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B.Tech – Electronics and Communication Engineering

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EDUCATION

Degree/Certificate	Institute/Board	CGPA/%	Year
B.Tech.	Indian Institute of Technology Guwahati	6.86 (Current)	2024 – Present
Senior Secondary	Telangana Board of Intermediate Education	96.2%	2024
Secondary	Telangana Board of Secondary Education	9.7	2022

PROJECTS

- **CVM++ — Stack-Based Virtual Machine & Compiler** Jan. 2026 – Apr. 2026
IIT Guwahati Coding Club — Even Semester Projects 2026 (Team) GitHub
 - Built a C++17 Lexer → Parser → Bytecode Compiler → Stack VM pipeline with zero external dependencies.
 - Designed a flat `uint8_t` bytecode format with 23 custom opcodes and $O(1)$ name-to-slot variable access.
 - Implemented 2-pass forward-jump patching resolving `OP_JMP` target offsets only after the full body is compiled.
 - Supported variables, arithmetic, booleans, control flow, and I/O via a fetch-decode-execute stack execution model.
 - Shipped `-debug` mode with an AST pretty-printer and bytecode disassembler annotating each opcode with hex offset.
- **MWIS Solver — Maximum Weight Independent Set** Jan. 2025 – Apr. 2025
IIT Guwahati Coding Club — Even Semester Projects 2025 (Team) GitHub
 - Built a 3-phase solver: branch-and-bound DFS with suffix-sum pruning over top-24 skill-to-degree ranked nodes.
 - Used 32-bit adjacency bitmasks for $O(1)$ conflict detection on graphs up to 200,000 nodes and $O(N^2)$ edges.
 - Applied `getchar_unlocked` buffered I/O achieving 10–100× faster input parsing than `cin`.
 - Boosted weight 10–50% over greedy via ADD/SWAP local search with $\pm 5\%$ perturbed restarts within 300s.
- **Parker — Personal AI Assistant with Persistent Memory** 2025 – Present
Personal Project (Ongoing) GitHub
 - Architected 4-layer memory: ranked fact store with 90–365-day archival thresholds and a multi-session project tracker.
 - Built a 5-level episodic summary tree (chat → day → week → month → year) stored in PostgreSQL.
 - Used 1,024-dim pgvector embeddings for semantic retrieval of facts and episodic summaries across all sessions.
 - Designed a LangGraph state machine (trigger → retrieve → chat → remember) with 4 isolated Groq API keys.
 - Deployed Telegram bot with text and Whisper voice input; injected profile, facts, and episodic context each turn.

TECHNICAL SKILLS

- **Programming:** C, C++, Python*
- **Web Technologies:** HTML, CSS, JavaScript, React*
- **Tools/Frameworks:** NumPy, Pandas, Matplotlib, SymPy, NetworkX*, Git*, GitHub*
- **Operating Systems:** Windows*, Linux * Elementary proficiency

ACHIEVEMENTS

- **Competitive Programming**, Max Rating **1404** on Codeforces (**Specialist**); Handle - `ssp_09` 2025 – 26
 - Ranked **1320** globally among 34,550+ participants in Codeforces Round 1090 (Div. 4).
 - Achieved Global Rank **1994** among 21,113+ participants in Codeforces Round 1083 (Div. 2).
- **JEE Advanced 2024**, Ranked in the top **2.53%** among **1.8 lakh+** shortlisted candidates across India 2024
- **TS EAPCET 2024**, Ranked in the top **0.87%** among **2.4 lakh+** candidates across Telangana State 2024

KEY COURSES TAKEN

- **Computer Science:** Introduction to Computing, Computing Laboratory
- **Mathematics:** Linear Algebra, Calculus, Differential Equations, Real Analysis, Complex Analysis
- **Core Engineering:** Basic Electronics, Digital Circuits (Theory & Lab), Circuit Theory, Signals and Systems

EXTRACURRICULAR ACTIVITIES

- **PR & Branding Associate**, Alcheringa — Northeast India's largest cultural fest Nov. 2024 – Feb. 2025
- **Junior Web Developer**, Students' Web Committee — frontend dev on institute projects Oct. 2025 – Present
- **HabitFlow**, Personal project — React habit tracker with streaks, analytics & progress visualization Oct. 2025
- **NSS Volunteer**, NSS Cell, IIT Guwahati — 30+ hours of community service and outreach activities Jan. 2026 – Present